

Each block node has a complete copy of the distributed ledger within a blockchain network. As illustrated in Figure 3, a valid block hash begins with four leading zeros, indicating its

validity. Each new hash is generated by combining the previous hash, the new transaction block, and a nonce. The difficulty of the hash, determined by the number of leading zeros it must have,

regulates the speed of block production to approximately one block every 10 minutes. A higher difficulty makes it more challenging for anyone to alter the blockchain and use it for double spending.

The image displays two side-by-side forms representing the process of mining a new block in a blockchain. Each form has a light green background and contains the following fields:

- Block:** A dropdown menu showing the block number (# 1 for the left form, # 2 for the right form).
- Nonce:** A text input field containing the nonce value (11316 for the left form, 35230 for the right form).
- Data:** A large text area for entering transaction data. In the left form, it contains a single character 'I' and a green circular icon with a white 'C' in the bottom right corner. In the right form, it is empty.
- Prev:** A text input field showing the previous block's hash. For Block #1, it is a string of 32 zeros. For Block #2, it is a long alphanumeric string.
- Hash:** A text input field showing the new block's hash. For Block #1, it starts with four zeros followed by a long alphanumeric string. For Block #2, it also starts with four zeros followed by a long alphanumeric string.
- Mine:** A blue button at the bottom of each form.

Figure 3: Demonstration of blockchain working on distributed nodes

Table 2: Popular blockchain platforms

| Criteria             | Ethereum                              | Hyperledger Fabric                                       | R3 Corda                                                    | Quorum                                                           |
|----------------------|---------------------------------------|----------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------|
| Type                 | Public                                | Permissioned                                             | Permissioned                                                | Permissioned                                                     |
| Target industry      | General-purpose blockchain platform   | Various, but primarily enterprise                        | Financial services, supply chain, healthcare                | Financial services, enterprise                                   |
| Programming built-in | Solidity                              | Go, Java, JavaScript                                     | Java, Kotlin                                                | Solidity, Go, Java                                               |
| Governance           | Ethereum developers                   | Linux Foundation                                         | R3 Consortium                                               | J.P. Morgan and network participants                             |
| Cryptocurrency used  | Ether                                 | Can integrate with various cryptocurrencies              | Not applicable (permissioned network)                       | Quorum ERC-20 tokens, but can incorporate other cryptocurrencies |
| Privacy feature      | Limited                               | Supported                                                | Supported                                                   | Supported                                                        |
| Consensus mechanism  | Proof of Work (PoW)                   | Pluggable framework                                      | Pluggable (supports various consensus algorithms)           | Pluggable (based on Ethereum)                                    |
| Complexity           | High complexity                       | Moderate-- requires understanding of Fabric architecture | Moderate-- requires understanding of Corda flows and states | Moderate--requires understanding of Ethereum-based architecture  |
| Transaction fees     | Gas fees vary based on network demand | Configurable by network operators                        | Not applicable (permissioned network)                       | Configurable by network administrators                           |

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